Did Generative AI Trigger a Structural Break? ARIMA Analysis of XLK ETF

1 Introduction

Technology stocks have accounted for 32% of global equity returns and 40% of total U.S. equity market returns since 2010 (Goldman Sachs, 2024), highlighting the dominant role of the technology sector in modern financial markets. Large-cap firms such as Apple and Microsoft represent a substantial share of total market capitalization, and their performance increasingly shapes the trajectory of major stock indices such as the S&P 500 and the NASDAQ Composite. Because technology companies are largely valued based on expectations of future innovation and scalability, new technological breakthroughs can trigger significant revisions in investor expectations, often leading to substantial movements in stock prices.

The invention of a new technology brings lots of impacts to a company's business. Especially for technology companies, launching a new technology product increases the valuation of company (McKinsey,2016).

As several Tech Giant firms launching their AI product, such as the Gemini from Google and the Claud from Anthropic, their companys' valuation, specifically the stock price have shifted. Even though OpenAI is not publicly traded, the hard launch of ChatGPT on November 2022 has brought up OpenAI's net worth and attracted a great amount of investors, with it's valuation skyrocketing from roughly $20 billion to an estimated 852 billion by 2026 (Barrientos, 2026).

The adoption of AI applications reshaped market narratives surrounding technological growth and productivity. Not only the company launching AI products, companies perceived as key beneficiaries of AI development, including NVIDIA Corporation and Microsoft Corporation, also experienced significant revaluations in the subsequent period. This shift shows that innovation shocks may influence not only short-term prices but potentially the broader dynamics of the technology sector.

Understanding whether such a technological breakthrough corresponds to a lasting change in market behavior is important for investors and policymakers. Persistent shifts in return

patterns could affect risk assessment, portfolio allocation, and forecasting models. Financial return series are time-dependent and often exhibit changes in volatility and growth dynamics over time. If the emergence of generative AI altered investor expectations in a sustained way, this effect may manifest as a structural change in the return-generating process.

The goal of the paper is to investigate whether the emergence of generative AI displays a change in the behavior of technology-sector stock return. The change is evaluated by comparing the stock prices before and after the ChatGPT launched time stamp. Specifically, we analyzes the Technology Selected Sector SPDR Fund (XLK) using ARIMA models and examines whether the release of ChatGPT corresponds to a significant change in the XLK price from 2017 to 2024.

2 Data

The XLK stock price we are looking at is collected by S&P Dow Jones Indices and State Street Global Advisory. S&P Dow Jones Indices is in charge of creating the company benchmark on entering the technology company stock pool; while the State Street Global Advisors selected particular stocks into the combination of the XLK stock. The data we are using is collected from the investing.com website, a financial data platform that provides public real-time market data directly from stock exchanges such as the New York Stock Exchange and NASDAQ. These two trading platforms provide a transparent record of the performance of the Technology Selected Sector Index. We are interested in the influence of the large-cap U.S. technology companies such as Microsoft, NVIDIA, and Apple whose market valuations and returns constitute the movement of the XLK fund.

The variables we are using includes:

- Price (Close): The final price at which the ETF traded when the stock exchange closed (typically 4:00 PM EST).
- Open: The price of the very first transaction when the market opened (9:30 AM EST).
- High & Low: The maximum and minimum prices reached during the trading session.
- Vol.: The total number of shares traded during the day (e.g., 15.90M means 15.9 million shares).
- Change: The percentage difference between the current day's closing price and the previous day's closing price.

The data we are using is collected daily from January 3, 2017, to March 28, 2026, which incorporates the AI outbreak time.

Variable Creation and Data Cleaning

We observed outliers that are double the price of the remaining data points. We also found that for most weekends, prices are carried over from the Friday of the same week. These outliers correspond to weekend days, and theey have no data for the variable, Vol. We decided to remove these outliers because they are not consistent with the general pattern, which they doubled the Friday price. Lastly, we split the dataset into two groups using November 30, 2022 (the release date of ChatGPT) as the cutoff.

Table 1: Summary Statistics of Stock Price

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mean_price	sd_price	min_price	max_price
69.91683	33.95005	24.4	152.07

Table 1 shows the mean, standard deviation, minimum and maximum of the stock price between 2017/1/3 to 2026/3/27. # Section 3. Analytic Methods

Transform Data

Figure 1 shows the stock price over time. The red dashed line indicates the time when ChatGPT was released. Overall, the stock price shows a general upward trend during the period shown.

Figure 2 indicates that the time series is not white noise, as the ACFs are out of the boundary and decay very slowly, which looks like a unit root process. Thus, we need to transform the data to make it stationary before model fitting. For stock data, taking the natural log is a common way to transform the data. The first difference of the log price series, known as the log return, is then used to remove the trend and produce a stationary series suitable for model estimation.

Figure 3 shows the log return over time after transformation. The log return fluctuates around zero, and the fluctuations are small for most of the time. However, there are a few large spikes. The biggest fluctuations occur around 2020 when the pandemic began. This suggests that the pandemic caused substantial uncertainty and sharp movements in the market.

Figure 3 suggests the mean is around zero, which suggests log return time series might be stationary. But Figure 4 shows the ACF of log return, where several spikes that are not within the bound. Therefore, an Augmented Dickey-Fuller (ADF) test was conducted to formally determine whether the series is stationary.

The ADF test has a p value of 0.01, suggesting log return is stationary.

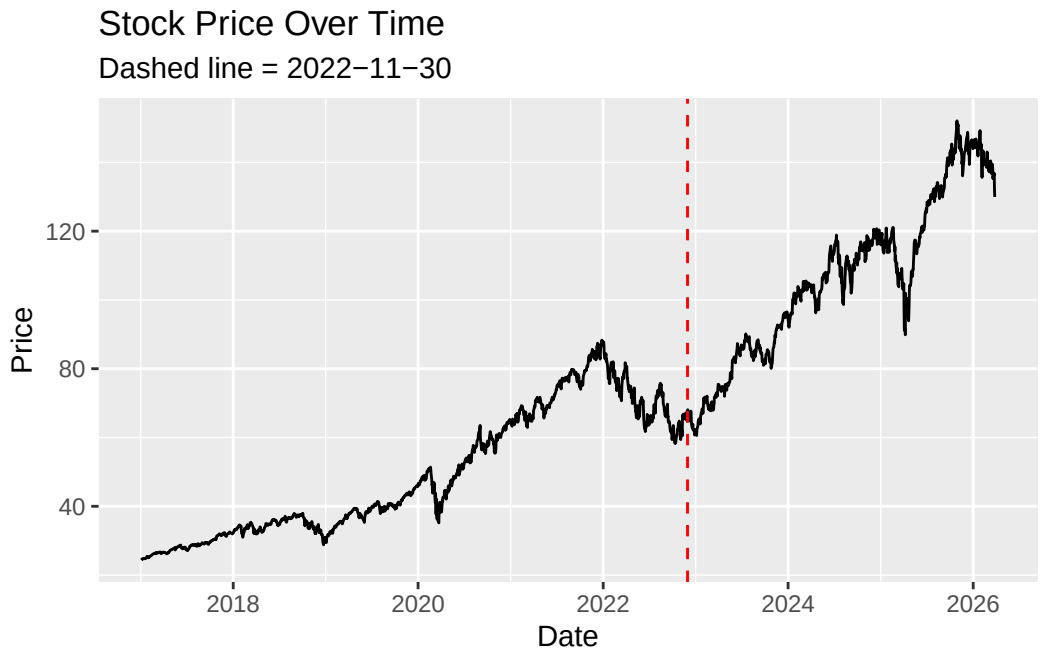


Figure 1

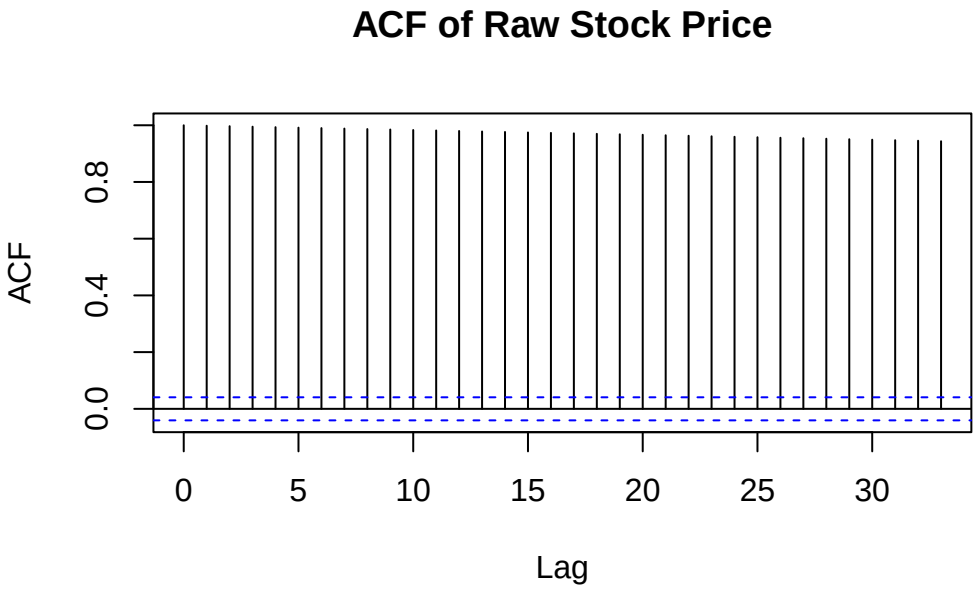


Figure 2

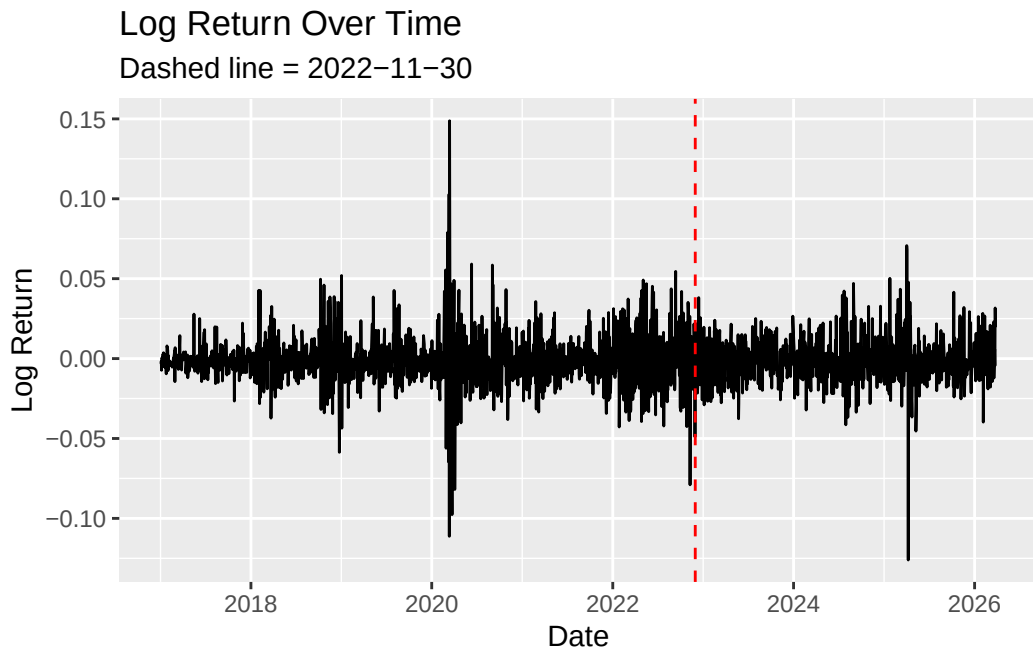


Figure 3

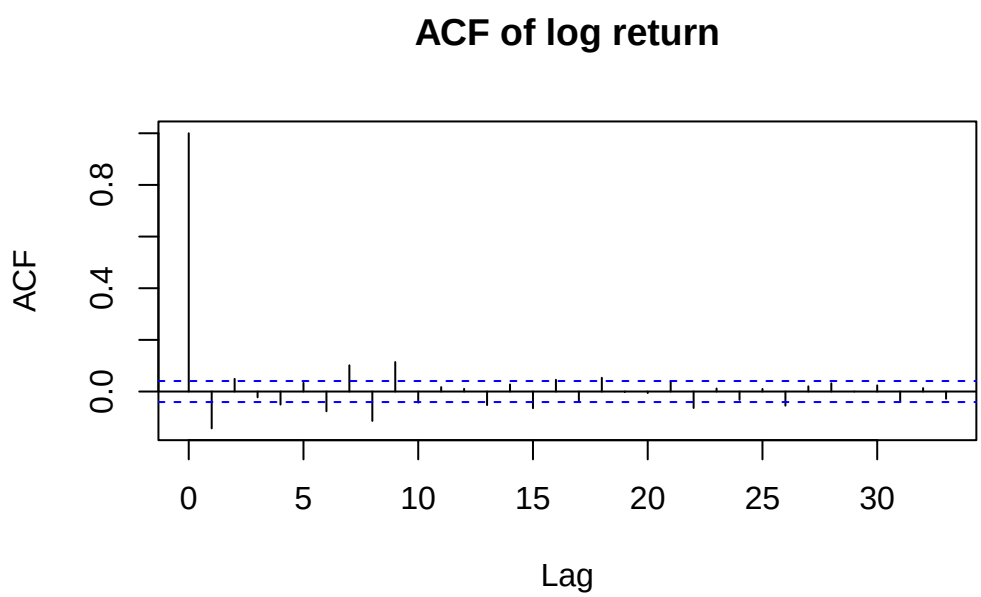


Figure 4

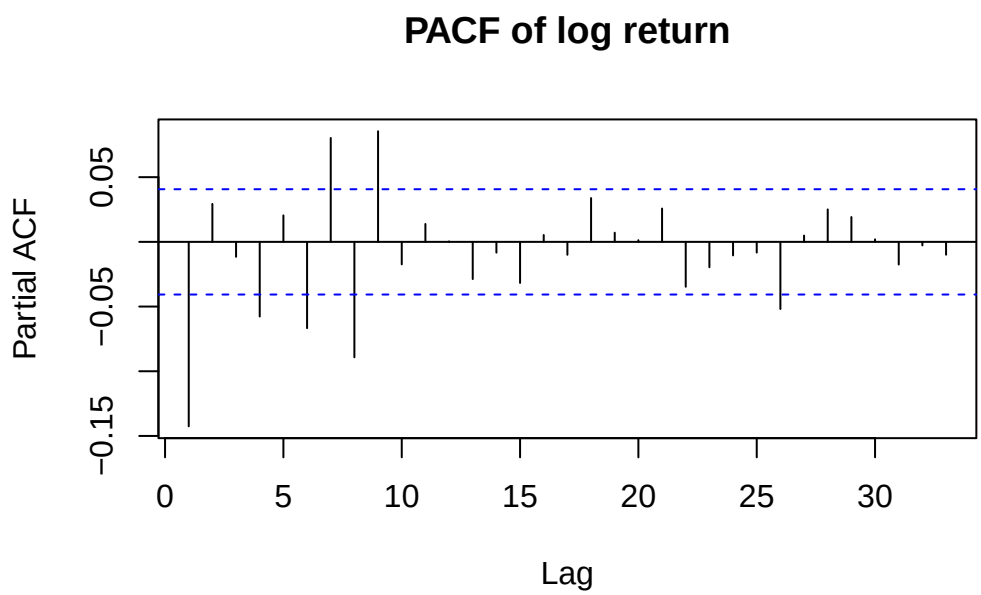


Figure 5

Compare log return using regression

Table 2: Summary Statistics of Stock Price

period	Mean	SD
After AI	-0.00078	0.01456
Before AI	-0.00069	0.01641

According to table Table 2, the mean log return before and after the appearance of ChatGPT is both close to zero. To test whether there is a difference on log return before and after ChatGPT appearance, a regression model with a dummy variable D_t as the predictor was fitted.

$$log(return) = \beta_0 + \beta_1 D_t$$

where $D_t = 0$ is before the appearance of ChatGPT, and $D_t = 1$ is after the appearance of ChatGPT. A two sample mean test can be used to determine whether D_t is significant. The null hypothesis is that the mean log returns in the two periods are equal, while the alternative hypothesis is that they are different.

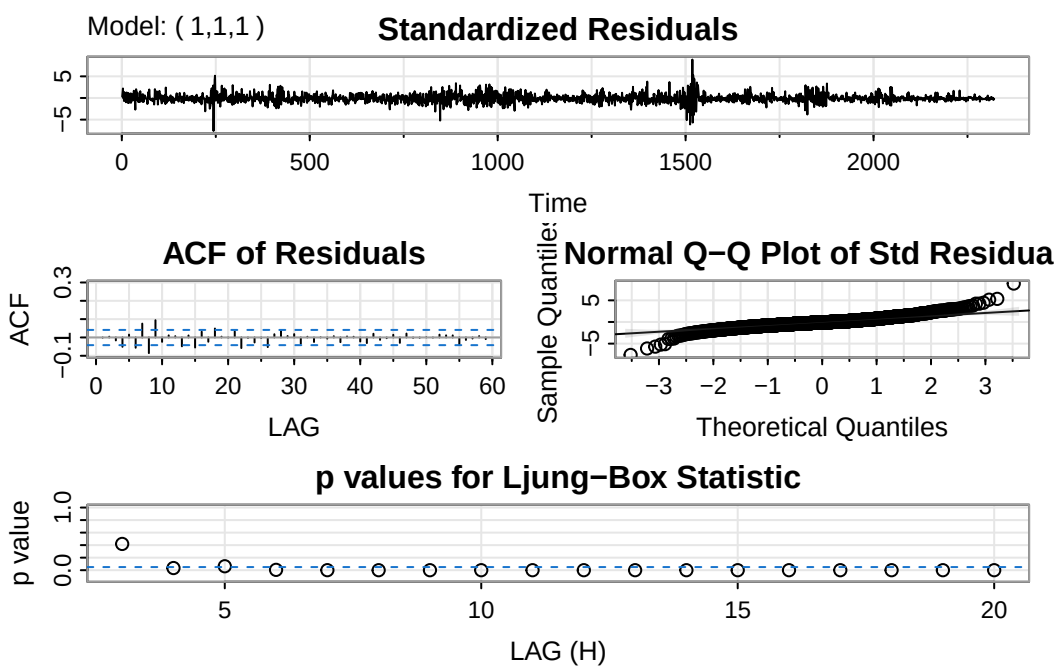
Table 3: Model Coefficients

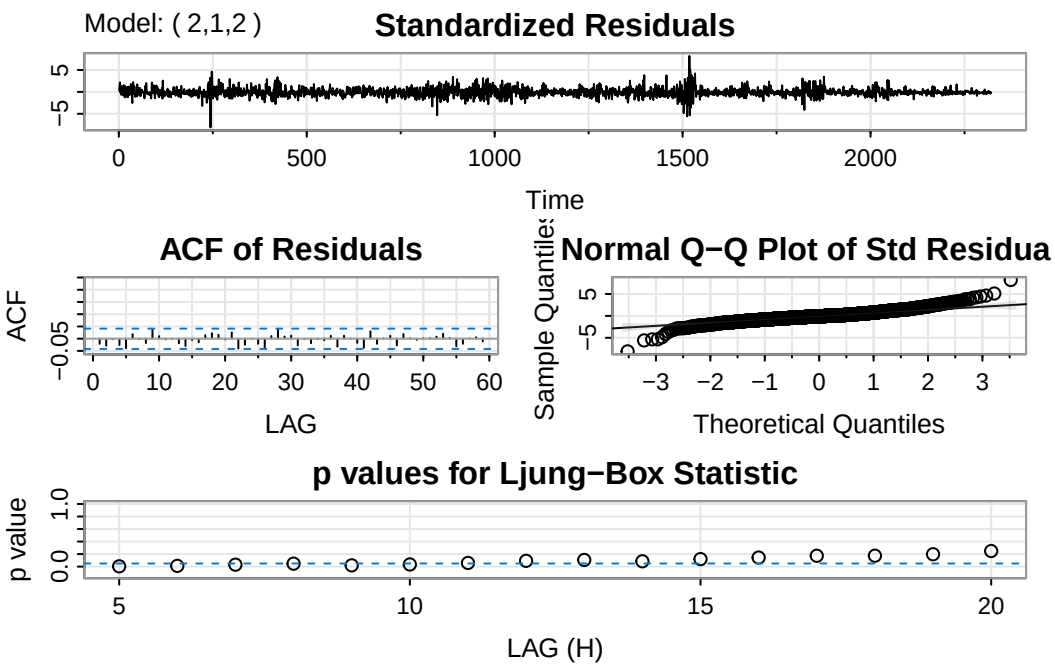
	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-0.001	0.000	-1.684	0.092
AI_dummy	0.000	0.001	-0.132	0.895

The p value of the dummy variable is 0.895, which is greater than 0.05. Thus, we fail to reject the null hypothesis, and conclude that there is no difference on the mean log return values before and after the appearance of AI.

Fit ARIMA Model

Using the ACF and PACF plots, we can determine the appropriate order of the ARIMA model. According to Figure 4, the ACF gradually tails off, suggesting the presence of autoregressive behavior. According to Figure 5, there are strong spikes at lags 2, 4, 7, and 8, and several other significant lags. Based on these patterns, ARIMA(1,1,1), ARIMA(2,1,2), ARIMA(4,1,4), ARIMA(7,1,7), and ARIMA(8,1,8) models were fitted to the price data to capture the time series.




Model: (2,1,2)
Standardized Residuals
Time
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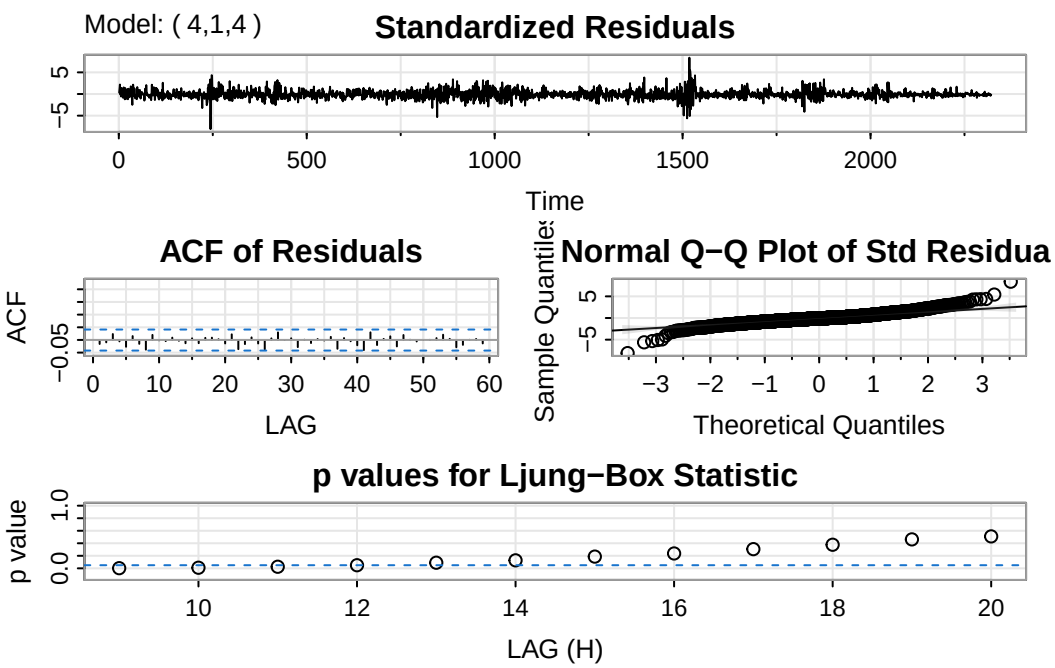
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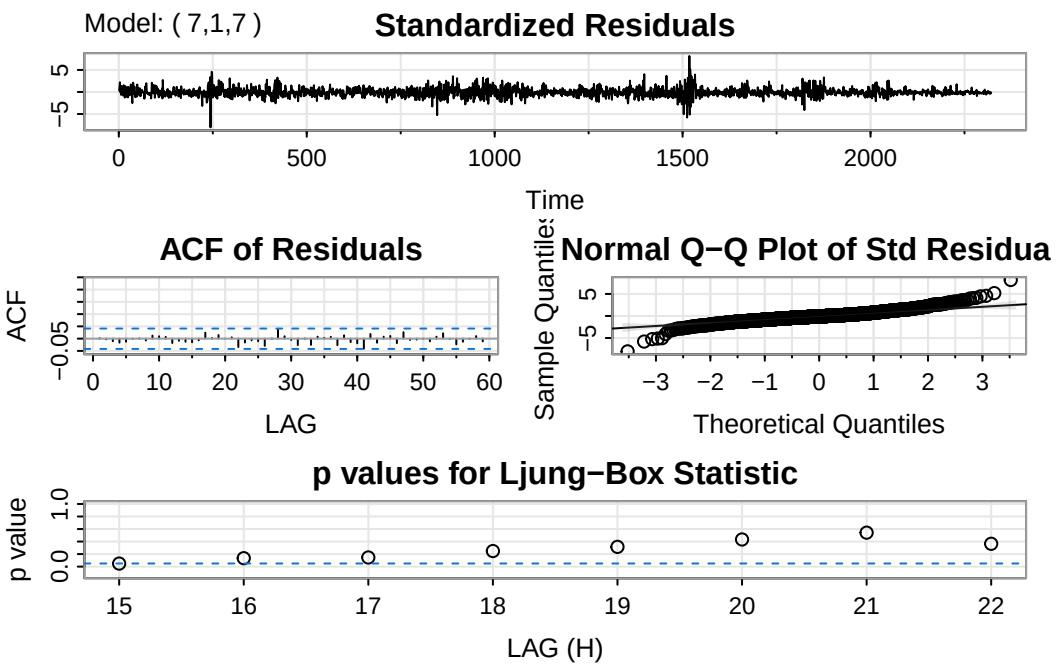
Model: (7,1,7)
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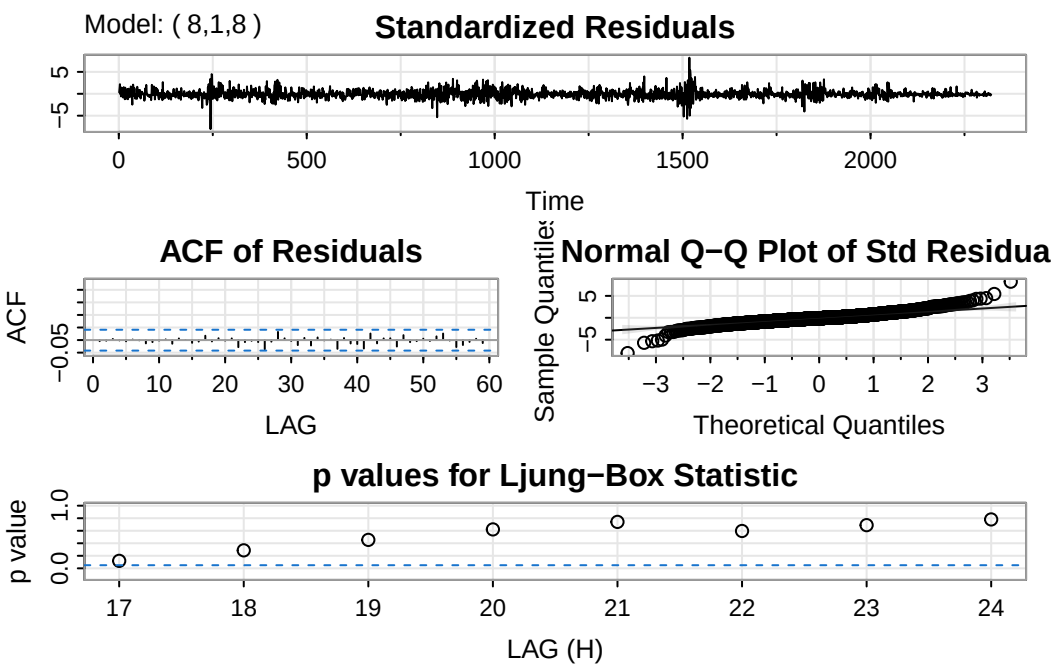
Model: (8,1,8)
Standardized Residuals
Time
ACF of Residuals
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LAG
Normal Q−Q Plot of Std Residua
Sample Quantiles
Theoretical Quantiles
p values for Ljung−Box Statistic
p value
LAG (H)

Model Diagnosis

The p values for Ljung-Box Statistic plots indicate that ARIMA(1,1,1), ARIMA(2,1,2), and ARIMA (4,1,4) are not good fits, because p values are significant for all lags. ACF plots for ARIMA(7,1,7) and ARIMA (8,1,8) looks stationary, but QQ plots show the points are deviated from the reference line, meaning the normality condition is not met. To determine which model performs the best, we compare the model AIC, AICc, and BIC values.

Model Choice

Table 4: Comparison of Model Performance

	AIC	AICc	BIC
ARIMA(1,1,1)	-5.477658	-5.477656	-5.470223
ARIMA(2,1,2)	-5.502680	-5.502673	-5.490290
ARIMA(4,1,4)	-5.503696	-5.503669	-5.481393
ARIMA(7,1,7)	-5.503852	-5.503774	-5.466680
ARIMA(8,1,8)	-5.501314	-5.501212	-5.459185

The ARIMA(7,1,7) has the smallest AICc value among all the models. Therefore, we pick ARIMA(7,1,7) as our best model.

Forcasting

We are interested in how significant the data after 2022/11/30, when ChatGPT got launched, is to forecasting future stock market prices. We will being using data frames of different time sets: 2017/1/3 - 2026/3/27, 2017/1/3 - 2022/11/30 and 2022/11/30 - 2026/3/27. (Date) The forecast models we will be using three of the "benchmark" methods: Mean method, Naive method and Seasonal naive method. (descibe methods) Drift method and decomposition method seem unnecessary with log return data because we have already removed the trend.

First, we aimed to examine whether AI has had an unexpected impact on the stock market and contributed to its fluctuations. We would like to use data from 2017/1/3 - 2022/11/30 to forecast data after 2022/11/30.

```r
library(tseries)

adf.test(log_return_ts)
```

```
Warning in adf.test(log_return_ts): p-value smaller than printed p-value
```

```
    Augmented Dickey-Fuller Test

data:  log_return_ts
Dickey-Fuller = -13.213, Lag order = 13, p-value = 0.01
alternative hypothesis: stationary
```

Table 5: Forecast Errors

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Method	MAE	RMSE
Mean	0.01049	0.01457
Naive	0.01241	0.01597
Seasonal	0.02161	0.02880

According to Table 5(explain meaning), all the scaled dependent errors are small, even for these easier approaches. It appears that the data prior to 2022/11/30 can effectively predict the data following the cutoff point. Furthermore, this suggests that AI have not yet contribute a significant impact in the stock market. As a result, we will use the entire time frame, allocating 80% for training and 20% for testing, to build a forecast model in predict prices in 2026.

Table 6: Forecast Errors-full model

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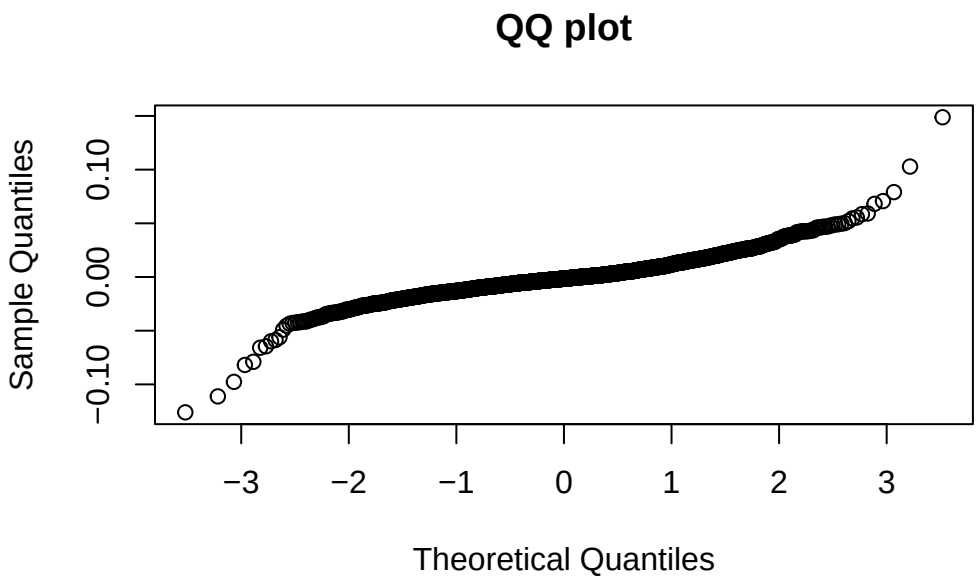
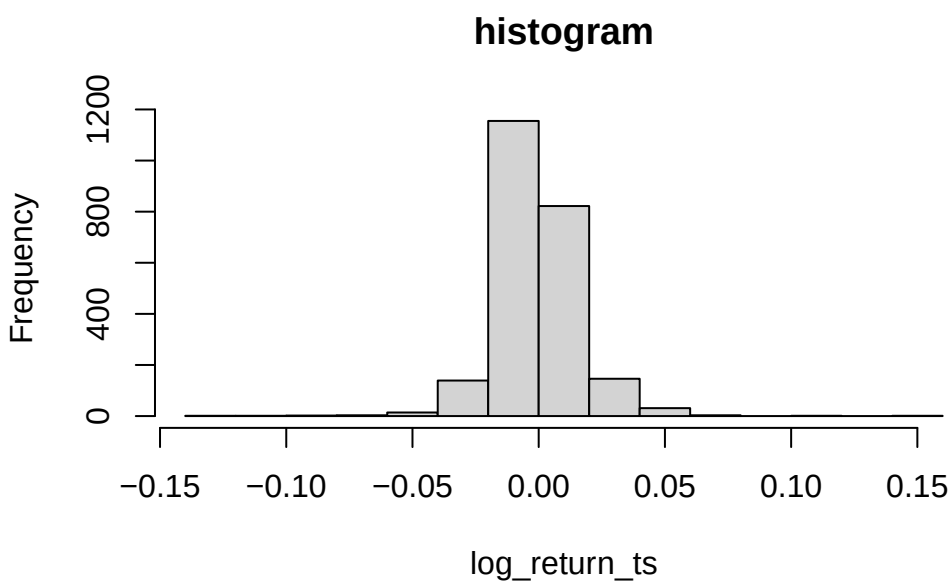
Method	MAE	RMSE
Mean	0.01128	0.01631
Naive	0.01459	0.01845
Seasonal	0.01311	0.01754

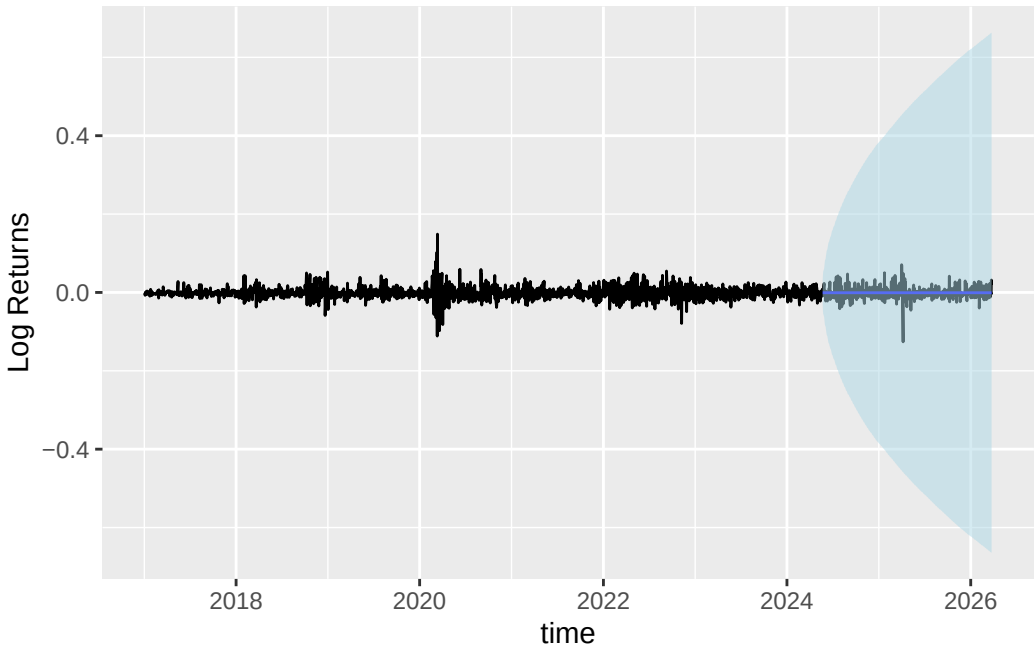
Table 6 shows the mean method is the best with lowest MAE and RMSE. We did not put MAPE here for comparison, because our log return data moves around 0, where some small values here make MAPE enormous and not as a good parameter to look at.





Next, we would like to also make a prediction interval for this forecast model.[ACF:( ]. The histogram looks normal, however qq-plot reveals that log return does not satisfy the linearity assumption. This suggests that our prediction interval is off.

(Will do ARIMA but also keeping this)



Summary

This project tests whether the rise of AI has significantly affected the behavior of technology sector stock returns. Using the XLK ETF price data from 2019 to 2026, we transformed the series into log returns to make the data stationary and applied AR(1), MA(1), and ARMA (1,1) time series models. The result suggests that stock return pattern remained relatively stable before and after the emergence of AI, indicating that AI has not yet had a significant impact on the technology stock market price. More broadly, this shows that while AI is often seen as a transformative force, its impact on financial markets may not be immediately reflected in stock return fluctuations.

Limitations

There are some potential limitations for the dataset. One of the limitations is that there's possibility for State Street to rebalance the stock composition over time, which means the dataset may not remain fully consistent throughout the study period. This could reduce the reliability of comparing stock performance across different years and weaken the robustness of analysis.

Another limitation is the dependency of the dataset. In Section 3, we found out the time series data points remain highly correlated even after the data transformation. For instance, the QQ plot indicates deviations from normality, suggesting that model assumptions may not hold

and prediction intervals may be unreliable. In addition, our model focuses on the short-term impacts after the release of ChatGPT, but the effect of AI may be better reflected in a long term trend rather than a boom. The log transformation may mask the underlying long term trend, because it removes the trend to make the data stationary. The log transformation only focuses on daily changes rather than long-term structural changes. As a results, model based on the log transformation is better at studying short-term market reaction but may be less effective at capturing long-term changes in the market over time.

Also, AR, MA, ARMA are all models that are only a function of the time series itself. They do not include other important factors, such as government policy or economic policy changes. There are many other factors that affects the stock market that we did not incorporate, such as economic indicators and interest rates. Since stock market is influenced by many external variables, excluding these factors may limit the strength and forecasting accuracy of the models.

Future Directions

To account for external factors that might affect the stock market, we could add these as predictors in our model. This can help us better estimate stock prices.

Although we have not been able to conclude AI has significant impact on the behavior of technology sector stock returns, it is still useful to estimate how sensitive technology sector stock price comparing to the broader market indices or other sectors. We might expect the technology sector to show higher volatility and respond more to growth expectations. In future studies, we can fit and compare separate models for different market segments, such as the S&P 500 and individual sectors like healthcare or financials, to better understand these differences.

It is also possible to incorporate the method proposed by Campos-Martins (2024) with our model. After identifying notable development by AI companies, we can adjust our predictors to improve accuracy. For example, we could include an indicator variable that captures major AI events or achievements. This would allow the model to account for changes over time driven by new innovations, instead of relying only on fixed predictors.

Finally, our research should follow a consistent process where we constantly refit the models as the stock market changes over time. Updating the model with new data can help us see whether there are any long-term effects or clear shifts in the data, especially as AI continues to develop and become more widely used.